LEONARDO **VICENTINI**

Software Engineer · Platform & Infrastructure

Open to relocation across the EU, UK & CH

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Skills

Programming	Go, Python, Java, JavaScript, Bash, SQL
Platform & Cloud	Kubernetes, Kubernetes Operators, OpenAPI, REST APIs, Docker, Helm, AWS, GCP, Azure, Linux
DevOps & MLOps	GitOps, Ansible, OpenTofu (Terraform), CI/CD, Prometheus, Grafana, MLflow, KServe

Education

University of Trento — *Trento, Italy*

Master's degree in Computer Science (English) — Grade: 110/110 with honors (GPA: 29.73/30) — *Sep. 2021 — Mar. 2025*

- Relevant courses: Distributed Systems, Cloud Computing, Service Design, Security Testing, Blockchain, Data Mining, HPC

Bachelor's degree in Computer Science — Grade: 106/110 — *Sep. 2018 — Sep. 2021*

- Relevant courses: Algorithms & Data Structures, Software Engineering, OOP, Operating Systems, Databases, Networks, HCI, ML

Work Experience

Krateo - Cloud platform startup supporting enterprise clients — *Verona, Italy (Remote)*

Software Engineer - Kubernetes, Cloud & Platform — *May 2025 — Jun. 2026*

- Owned the end-to-end development of core components of the Kubernetes-based Krateo platform, from design through production.
- Built a Postgres-backed query API in Go with RBAC-aware filtering and keyset pagination, eliminating slow on-demand resource queries that degraded frontend UX, cutting latency from ∼7s to ∼400ms and sustaining 270+ req/s, validated via 150+ load-test runs.
- Evolved a Kubernetes Operator framework that auto-generates CRDs and controllers from OpenAPI specs, eliminating hand-written operator code for APIs not natively managed in Kubernetes and isolating failures per resource type.
- Independently migrated and extended 2 legacy integrations (Azure DevOps, GitHub) onto the OpenAPI-driven operator framework, growing coverage to 20 production resource types published to the platform marketplace.
- Provided rotational L1–L3 support for 4 enterprise deployments spanning 12 Kubernetes clusters (dev/test/prod), diagnosing incidents and performing root cause analysis across clusters, platform components, and external integrations.
- Designed, documented, and oversaw rollout of near-zero-downtime platform upgrades across 4 enterprise deployments, validated against replicated client environments and codified into reusable runbooks.
- Evaluated frameworks (kagent, Google ADK), memory systems, and MCP/A2A protocols for Krateo Autopilot, a multi-agent system.

Master thesis researcher - Kubernetes, Cloud & Platform — [**Journal Paper**] — *Oct. 2024 — Apr. 2025*

- Integrated Open Policy Agent as a Kubernetes mutating webhook to enforce latency-SLA, GDPR data-residency, and carbon-aware scheduling policies, contributing to a system achieving up to 13% CO2 emission reductions vs. round-robin baseline scheduling.
- Developed Helm templates that dynamically select the best-fit VM size, gate provisioning on cloud provider and scheduling time, and drive VM and networking infrastructure creation across AWS, GCP, and Azure via native Kubernetes Operators.
- Architected and deployed an MLOps pipeline serving ∼15 PyTorch carbon-forecasting models using MLflow and KServe on K8s.

ESA – European Space Agency (European Space Operations Centre) — *Darmstadt, Germany*

Software Engineer Intern - Cloud & Infrastructure — *Apr. 2024 — Jul. 2024*

- Delivered a GitOps-style Service Status Board with a GitLab CI/CD pipeline enforcing schema validation on incident, intervention, and announcement metadata before publishing, used by 3 Space Mission Operations teams to track critical infrastructure systems.
- Designed a Prometheus-based instrumentation layer with self-registering targets, spanning 5 exporter types (OS, Windows, process, network/SNMP, and uptime checks) across 10+ infrastructure sources and 4 Linux distributions.
- Built 15 Grafana dashboards visualizing time-series infrastructure metrics and an Alertmanager notification pipeline via Telegram, delivering near real-time health visibility for space mission operations.
- Automated secure deployment and TLS provisioning for monitoring agents via Ansible across 10s of VMs in restricted Operations LAN.
- Deployed 3 production instances of the monitoring platform using Kubernetes and Helm following a cloud-native architecture.

FIPIC – Italian Wheelchair Basketball Federation — *Rome, Italy (Remote)*

Software Engineer Intern - Backend — *Feb. 2021 — Jun. 2021*

- Led a team of 4 engineers to develop a federation-wide historical data and multimedia archive, reducing delivery time by ∼50% by improving planning, task distribution, and development workflows.
- Built an ELK-based data pipeline powering 4 dashboards, and exposed 70+ Node.js REST endpoints over a MySQL database.

Projects

systemslab — Deterministic Systems Simulator with Fault Injection

Go, React, TypeScript, OpenTofu, Podman — [**Live site: systemslab.dev**]

- Designed and built a deterministic discrete-event simulator in Go with 100+ scenarios across Raft, Paxos, 2PC, system design (retries, caching), LLM serving and disaster recovery, making runs reproducible from spec and seed so A/B diffs pinpoint divergence.
- Deployed it to production: OpenTofu-provisioned Hetzner VM reachable only through Cloudflare, Podman releases promoted by image digest with health-gated automatic rollback in ~2 min, per-client rate limits and a spend guard capping monthly cost.